Application note: TDbasedUFE and TDbasedUFEadv: bioconductor packages to perform tensor decomposition based unsupervised feature extraction

Overview

- We have developed the method called "Tensor decomposition (TD) based unsupervised feature extraction (FE)".
- It can perform unsupervised feature selection in the situation of "*large p small n*" problem.
- Two bioconductor packages by which one can easily perform TD based unsupervised FE were released as Bioconductor packages.

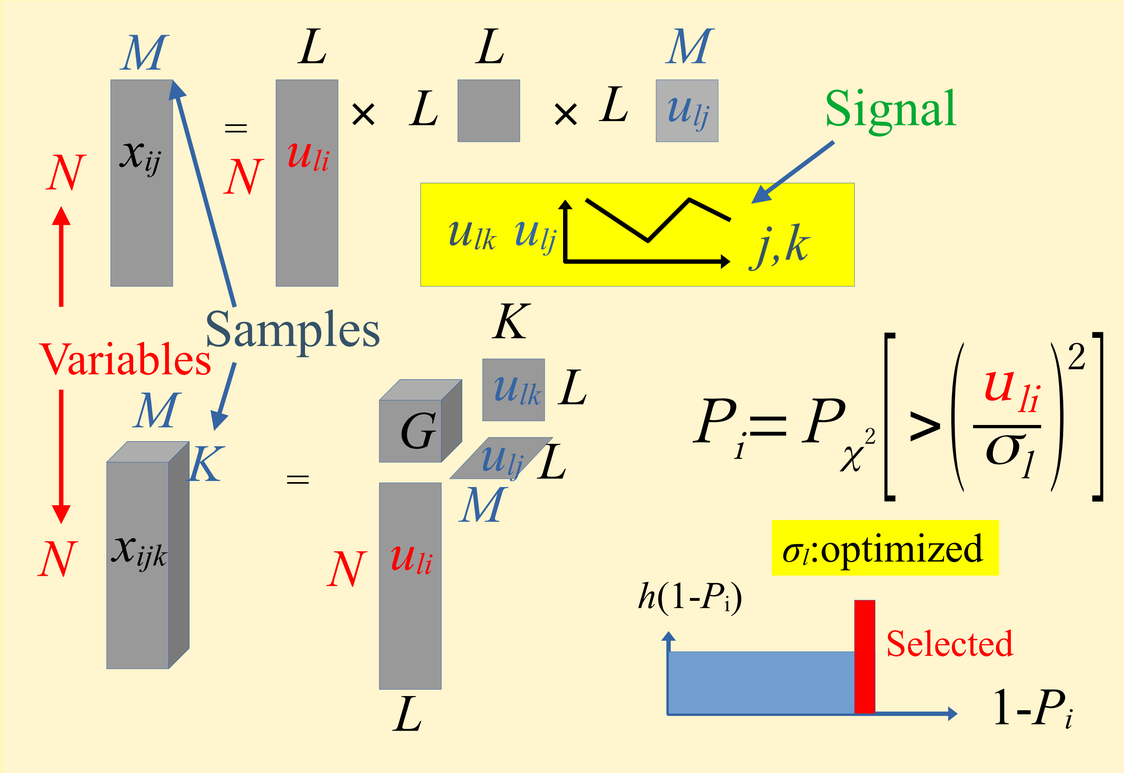
Introduction

So called "*large p small n*" is usual in genomic studies, since there are usually small number of samples (because of e.g., high cost or rare diseases) associated with large number of features (e.g., genes ~ 10^4 or genomic sites ~ 10^7). Although there are huge number of methods invented to perform feature selection (e.g., the identification of differentially expressed genes), there remain several difficulties in these methods, e. g., not filly adapted to multi-omics analysis, unstable selection, batch effect, and support for missing values. During a decade, I developed the method called "principal component analysis (PCA)/tensor decomposition (TD) based unsupervised feature extraction" to attack this problem. Last year we have released two bioconductor packages to easily perform this method and 500 pages 2nd Edition of the 1st edition of 300 pages length to fully describe the methodology published in 2019 will be published in this August from Springer. In this occasion, I come here to advertise my methodology which is not yet popular at all.

Advantages of TD based unsupervised FE

- Missing values need not to be filled since TD can fill them automatically.
- No treatments are required for batch effects since batch effects can be separated as independent components.
- The stable feature selections is possible even in "*large p small n*" situation.
- Very fast, since no learning/optimization is required.

Selecting variables from tensors/matrices



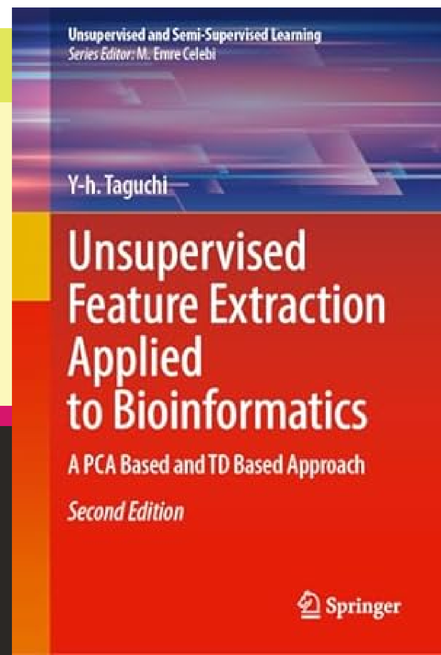
List of topics treated by the method

From the table of contents of the book.

- Biomarker identification using circulating miRNA
- Understanding soldier's heart from the mRNA and miRNA
- Identification of genes mediating transgenerational epigenetics based upon integrated analysis of mRNA expression and promoter methylation
- Single-cell multiomics data analysis
- Integrated analysis of PPI and gene expression
- Integrated analysis of epitranscriptome and mRNA expression
- Drug repositioning for SARS-CoV-2
- Gene expression and methylation in social insects
- Integrated analysis of mRNA and miRNA
- Drug-drug interaction detection based on gene expression profiles
- Histone modification bookmarks in post-mitotic transcriptional reactivation
- One-class differential expression analysis for multiomics data set
- Identification of genes mediating transgenerational epigenetics based upon integrated analysis of mRNA expression and promoter methylation

Conclusions

- The book about TD based unsupervised FE is published (2024 Aug., 2nd Ed.)
- The original paper of the poster: Taguchi & Turki (2023), https://doi.org/10.3389/frai.2023.1237542
- There are huge number of applications to real examples.
- Biocondutor packages: TDbasedUFE https://doi.org/doi:10.18129/B9.bioc.TDbasedUFE
- TDbasedUFEadv https://doi.org/doi:10.18129/B9.bioc.TDbasedUFEadv



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